Panel MS-AR(1) with random-effects intercepts, 1970Q1 onward

Pulled real GDP per worker

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g\,t + z_{it}, \tag{1}$$
$$z_{i,t+1} = \big(1 - \rho(s_{it})\big)\,\mu(s_{it}) + \rho(s_{it})\,z_{it} + \sigma(s_{it})\,\varepsilon_{it}, \tag{2}$$
$$s_{i,t+1} \sim \Pi(\,\cdot \mid s_{it}), \tag{3}$$
$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad \text{(random effects)}. \tag{4}$$

The outcome is log real GDP per worker from the pulled quarterly panel (constant 2015 USD, market FX). The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles are plotted using the posterior mean of a_i. Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1}. σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.995$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π. $E[z]$ is the long-run mean of the cycle implied by μ, ρ, and Π.

2 Common intercept and linear trend

Pulled panel from 1970Q1: log real GDP per worker. 101 countries, 7354 observations, 1970-Q1–2026-Q2. Fitted countries: 92. Observations: 6787. Log-likelihood: 14863.42. Ergodic mean of the cycle $E[z] = -0.0985$ (0.1273). Common intercept $a = 10.4145$ (0.1330). Common slope $g = 0.0037$ (0.0010).

Table 1: Regime AR(1), transition matrix, and ergodic π. Columns are regimes.

	(0)	(1)	(2)
μ	-1.2653	0.0000	0.8509
	(0.1601)	—	(0.0994)
σ	0.1217	0.0272	0.0111
	(0.0032)	(0.0006)	(0.0002)
ρ	0.9905	0.9950	0.9950
	(0.0020)	(0.0000)	(0.0000)
$\Pi_{0\cdot}$	0.9328	0.0570	0.0102
	(0.0095)	(0.0092)	(0.0042)
$\Pi_{1\cdot}$	0.0175	0.9591	0.0234
	(0.0034)	(0.0048)	(0.0041)
$\Pi_{2\cdot}$	0.0147	0.0157	0.9697
	(0.0032)	(0.0040)	(0.0037)
π	0.1937	0.4187	0.3876
	(0.0227)	(0.0305)	(0.0330)

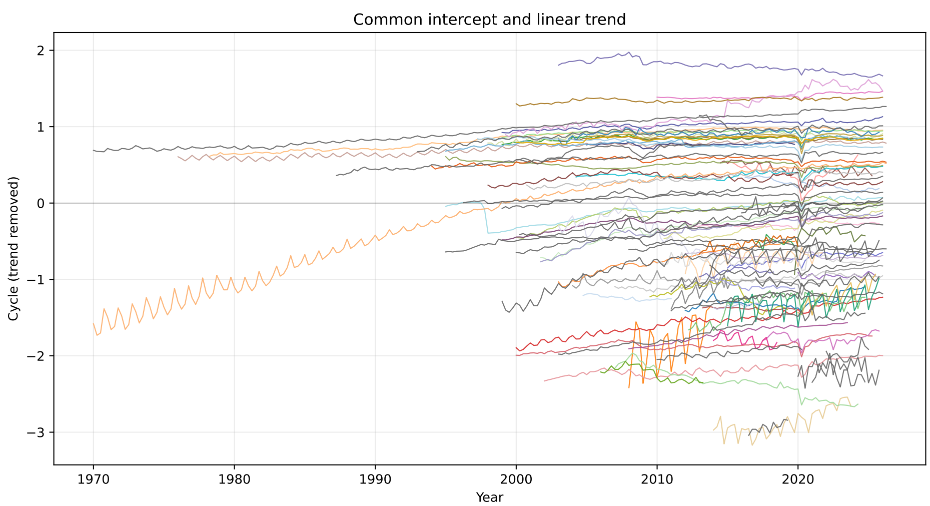

Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

Pulled panel from 1970Q1: log real GDP per worker. 101 countries, 7354 observations, 1970-Q1–2026-Q2. Fitted countries: 92. Observations: 6787. Log-likelihood: 15090.50. Ergodic mean of the cycle $E[z] = -0.1636$ (0.1083). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = 9.8429$ (0.0823), $\omega = 0.7204$ (0.0238). Posterior-mean a_i: mean 9.845, min 7.996, max 11.684. Common slope $g = 0.0084$ (0.0008).

Table 2: Regime AR(1), transition matrix, and ergodic π. Columns are regimes.

	(0)	(1)	(2)
μ	-0.8152	0.0000	0.5277
	(0.1955)	—	(0.0827)
σ	0.1244	0.0271	0.0109
	(0.0032)	(0.0006)	(0.0002)
ρ	0.9804	0.9950	0.9950
	(0.0081)	(0.0002)	(0.0000)
$\Pi_{0\cdot}$	0.9351	0.0490	0.0159
	(0.0096)	(0.0089)	(0.0051)
$\Pi_{1\cdot}$	0.0123	0.9700	0.0177
	(0.0027)	(0.0041)	(0.0033)
$\Pi_{2\cdot}$	0.0125	0.0126	0.9749
	(0.0028)	(0.0033)	(0.0033)
π	0.1605	0.4323	0.4071
	(0.0209)	(0.0329)	(0.0339)

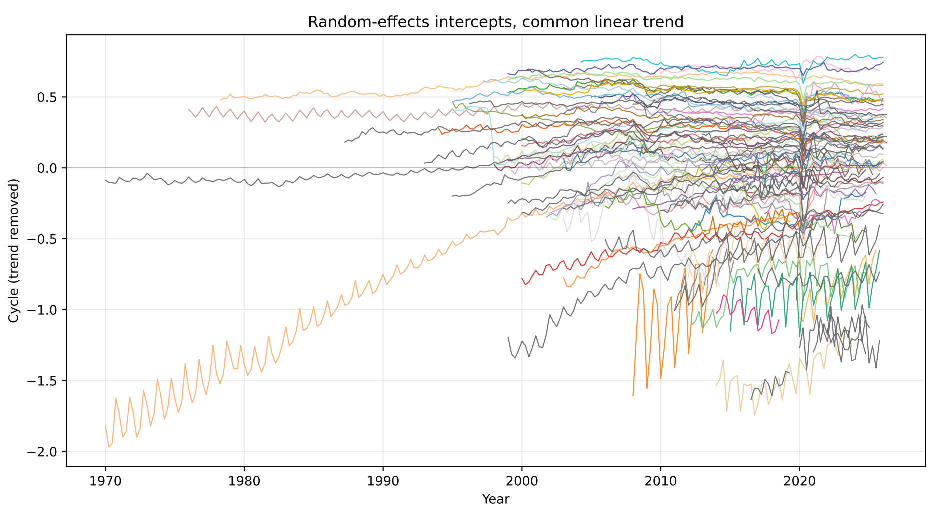

Figure 2: Country cycles after removing the linear trend.